

29.9% of returns aren't genuine. They carry **\$4.64M** of refund dollars — and now we know which ones.

A retailer's return-audit team reviews every disputed return the same way — spreadsheet speed, inconsistent criteria. We rebuilt **60,000 orders** into a PySpark/MLlib pipeline that triages each one into legitimate, policy abuse, fraud, or wardrobing, priced what's actually at stake, and separately segmented customers to surface the product-quality problem no fraud label can see.

NON-LEGITIMATE REFUND EXPOSURE

\$4.64M

43.8% of all refund dollars requested, from just **29.9%** of orders — bad actors skew toward higher-value claims

CLASSIFIER PERFORMANCE (TEST SET)

0.999

weighted F1, Random Forest — synthetic-dataset caveat on slide 7, not a production guarantee

HIDDEN IN THE "LEGITIMATE" MAJORITY

38%

of returns in the largest, 96%-legitimate customer segment are still quality-issue driven

60,000 orders · 35 features, Kaggle e-commerce return-abuse dataset · Fraudulent Return claims average **\$434.73** vs. **\$141.14** for legitimate returns — a **3.1x** gap · two independent PySpark MLlib approaches: multi-class classification (Random Forest vs. Logistic Regression) and K-Means clustering · this dataset is synthetic, so treat 0.999 F1 as a pipeline proof, not a production number (more on slide 7).

One review queue, *three different problems* wearing the same form.

A return-audit team currently reviews every disputed return the same way. It has no systematic way to tell whether a spike in returns is fraud, abuse of a lenient policy, or a genuine quality defect — three problems that need three different owners, currently indistinguishable in a spreadsheet.

01 · CUSTOMERS RETURN

Most returns are genuine

Defective items, wrong fit, changed mind — the normal cost of running an e-commerce store.

70.1% legitimate

02 · SOME ABUSE IT

Policy abuse, fraud, wardrobing

Exploiting lenient terms, deceptive claims, or wearing-and-returning — three distinct behaviors.

29.9% of orders

03 · ALL LOOK THE SAME

One manual-review process

Every disputed return gets identical treatment today, regardless of which of the three it actually is.

1 review queue

04 · THE WRONG FIX

Blanket policy tightening

Punishes the 70% of legitimate customers to catch the fraction who abuse the system.

wrong lever

THE QUESTION THIS PROJECT ANSWERS

Which returns are fraud, which are policy abuse, and which are genuine quality failures — and what does each one actually cost?

Three sub-problems, one dataset: **return fraud** → classification on abuse_type; **product quality** → return-reason patterns cut by cluster and category; **return risk** → lifetime order/return history per customer.

WHY THIS IS A TRIAGE PROBLEM, NOT A POLICY PROBLEM

The highest-value lever is routing each return to the right team fast — not tightening policy across the board, which would punish the 70% of customers who did nothing wrong. **Three different interventions, three different owners**, is the recommendation this whole pipeline builds toward.

INDUSTRY BACKGROUND

This is not a hypothetical problem — *the industry is already living it.*

E-commerce carries a structurally higher return burden than brick-and-mortar, and fraud/abuse is a large, growing share of it. This project sits squarely inside an industry-wide shift already underway.

NRF · 2025 RETAIL RETURNS LANDSCAPE

19.3%

of online sales are returned — above the 15.8% all-channel average. \$849.9B in merchandise returns forecast for 2025.

APPRISS RETAIL × DELOITTE · 2024

15.14%

of all returns are fraudulent — \$103B lost in 2024 (up from \$101B in 2023), across 60+ retailers studied.

SAME 2024 STUDY

60%

of retail executives specifically flag **wardrobing** — one of our four target classes — as a major fraud pattern. 85% now deploy AI to fight it.

HOW OUR DATASET COMPARES

This dataset's combined Fraudulent Return + Wardrobing rate is **10.2% + 7.7% = 17.9%** — close to Appriss's industry-wide **15.14%** fraud rate. The class **prevalence** here is realistically calibrated to real-world reporting, even though (see slide 7) the class **separability** the model achieves is not something to expect on live data.

Sources: National Retail Federation, *2025 Retail Returns Landscape* (nrf.com) · Appriss Retail with Deloitte, *Fraudulent Returns and Claims Cost Retailers \$103B in 2024* (apprissretail.com). Full citations in the report's Appendix C.

A realistic imbalance — and the dollars don't split the way the order counts do.



■ Legitimate — 70.1% (42,060 orders, \$5.94M requested) ■ Policy Abuser — 12.0% (7,192, \$0.76M) ■ Fraudulent Return — 10.2% (6,112, \$2.66M) ■ Wardrobing — 7.7% (4,636, \$1.22M)

Behavioral signals

Photo evidence, tracking validity, address changes, refund-to-different-account, multiple accounts, dispute history

Customer history

Lifetime orders/returns, return rate, account age, loyalty segment

Context

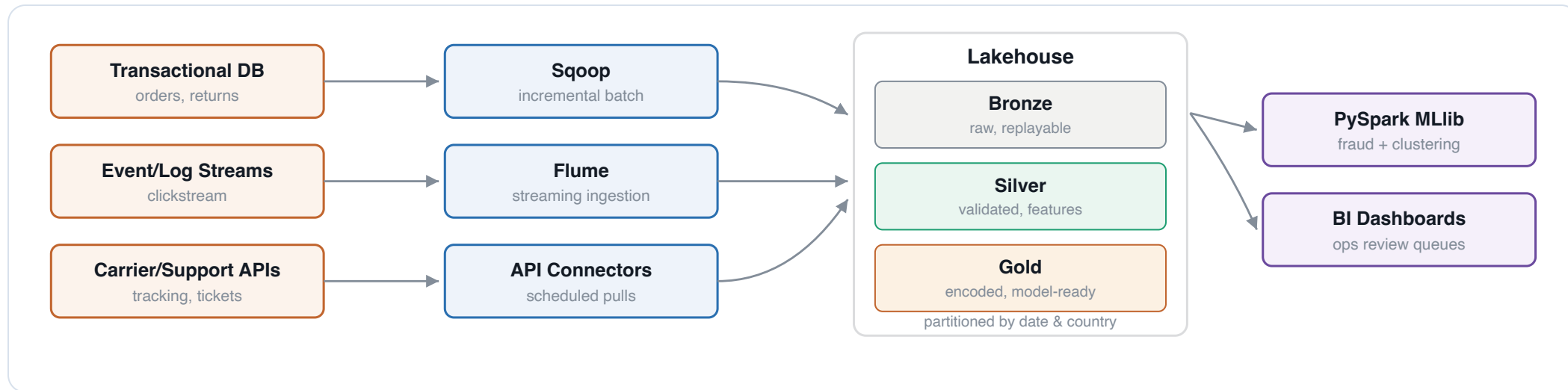
12 product categories, 12 return reasons, 8 countries, payment method, shipping carrier

VERIFIED — CHECKED PROGRAMMATICALLY, NOT ASSUMED

0 missing values across all 35 columns, **0 duplicate** order_id, **0 logical violations** (no return-before-order dates, no returns exceeding lifetime orders, no implausible ages) — confirmed in notebooks/01_data_engineering.ipynb §2 before anything downstream trusted this data.

Sqoop / Flume / API ingestion → *Bronze/Silver/Gold lakehouse* → PySpark MLlib.

This notebook runs directly on the static Kaggle CSV as a practical stand-in — the diagram below is the production-scale design the Silver→Gold transformation logic is built to generalize to.



WHY BRONZE/SILVER/GOLD

Bronze keeps a permanent, replayable raw record. Silver is cleaned and feature-engineered once, reused by both models. Gold is the only layer anything ML touches.

EXPLICIT SCHEMA, NOT INFERRED

The notebook loads the CSV with an explicit StructType — never inferSchema in production, which forces a full extra scan and can silently mis-type columns on drift.

CAP OUTLIERS, DON'T DELETE

Extreme values are often the fraud signal itself. 8 continuous fields IQR-capped at the 1st/99th percentile — preserving row count and the extremeness of the behavior.

What the three models actually tell us — *and the risks that travel with them.*

KEY FINDINGS

- **Non-legit returns skew expensive, not just frequent.** 29.9% of orders carry 43.8% of refund dollars (\$4.64M) — fraud claims average 3.1× a legitimate return.
- **The classifier earns its complexity.** Random Forest hits perfect precision/recall where a free 4-flag heuristic tops out at F1 0.61.
- **The quality problem is broad-based, not one vendor.** Quality-issue rates band 31.5%–35.2% across all 12 categories — a systemic pattern, not a vendor hunt.
- **A simpler model wins honestly on refund pricing.** Linear Regression beats both a naive baseline (40% RMSE) and a more complex GBTRegressor.
- **Our dataset's fraud rate is industry-realistic.** 17.9% (Fraud + Wardrobing) sits close to Appriss Retail's reported 15.14% — the scale here is real.

RISKS & FUTURE ROADMAP

- **Risk — synthetic-data overfit.** Treat this model as a pipeline proof-of-concept; re-train on real labeled outcomes before go-live, budgeting for 80–90% F1, not 99.9%.
- **Risk — no production ground-truth labels.** Real retailers rarely have a clean `abuse_type` column — build human-in-the-loop labeling from the review queue this system creates.
- **Risk — the quality finding explains *what*, not *why*.** Broad-based isn't root-caused; that needs a packaging/supplier QA audit this dataset can't run.
- **Roadmap.** Real-time scoring at return-request time, a feedback loop from reviewer overrides back into re-training, and a rolling time-based validation window once live data exists.

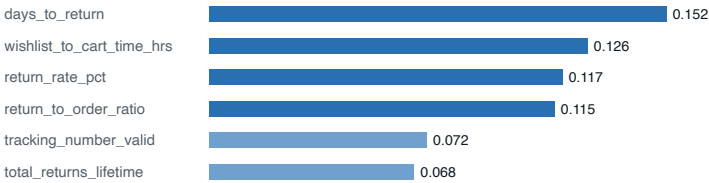
Random Forest vs. Logistic Regression on *return-fraud type*.

Spark MLlib's `GBTClassifier` only supports binary classification — not usable for this 4-class target — so Random Forest, which natively supports multi-class, is the primary model. Class-weighted training (inverse frequency) handles the 70/12/10/8 imbalance; evaluation reports per-class precision/recall/F1, not headline accuracy.

MODEL	WEIGHTED PRECISION	WEIGHTED RECALL	WEIGHTED F1
Logistic Regression (baseline)	0.998	0.998	0.998
Random Forest (final)	0.999	0.999	0.999

200 trees, maxDepth=10, class-weighted. Train 45,112 / test 14,888 rows, 75/25 split, seed=42.

TOP FEATURE IMPORTANCES



Engineered ratios rank alongside raw behavioral fields — the payoff of Phase 2.

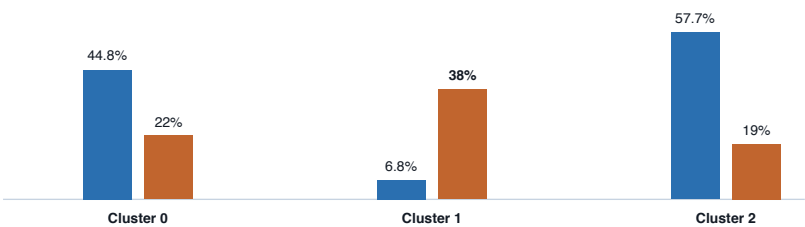
Honest note: this is a Kaggle synthetic dataset, so 0.999 F1 is higher than we'd expect in production — feature importance is spread across ~15 features with no single leak, but a realistic target on live data is **80–90% F1**. We treat this model as proof of the pipeline, not a performance guarantee. Full discussion in `docs/fraud-classification-results.md`.

APPROACH 2 — CLUSTERING

K-Means finds 3 behavioral segments *the fraud label can't see.*

An independent, unsupervised lens: features deliberately exclude the fraud-flag columns used by the classifier, so clustering surfaces behavior/quality patterns a supervised label alone can't. k=3 selected by silhouette score over k=2..6.

CLUSTER	N	AVG RETURN RATE	AVG ORDER VALUE	% QUALITY-ISSUEDOMINANT MIX
0 — High-value fraud-risk	6,646	44.8%	\$484	22% Fraudulent Return-heavy
1 — Core legitimate base	43,829	6.8%	\$157	38% 96% Legitimate
2 — Frequent policy-abuse	9,525	57.7%	\$129	19% Policy Abuser-heavy



KEY INSIGHT

Cluster 1 has by far the **lowest** return rate but the **highest** quality-issue share — the two metrics move in opposite directions, which is exactly why a single blended "return rate" KPI hides this segment's problem from whoever owns it today.

This is a **QA/vendor problem**, not a fraud problem — cross-reference dominant product_category within Cluster 1 against return_reason to target the escalation.

Four interventions, four owners — *ship the free one first.*

SHIP THIS FIRST, NO ML REQUIRED

Require photo evidence above a value threshold. The "high-value item + no photo evidence" pattern alone touches **\$1.88M** of non-legitimate refund exposure across **4,673 orders** — a policy rule, live immediately, independent of the classifier's rollout timeline.

01 Deploy the fraud classifier as a triage score, not an auto-block.

Route the highest fraud_signal_score decile + Cluster 0 to manual review; auto-approve the lowest-risk tail. **Re-threshold asymmetrically** — a false "Legitimate → Fraudulent Return" flag costs more in trust than a slow review costs in ops time.

OWNER · FRAUD & RISK

02 Give Cluster 1's 38% quality-issue rate to QA as a systemic audit, not a vendor hunt.

Quality-issue rates are broad-based — banded 31.5%–35.2% across all 12 categories, not one bad vendor. **73% of customers**, near-zero fraud, but the highest quality complaint rate of any segment.

OWNER · QA / VENDOR MANAGEMENT

03 Tighten policy — don't investigate — for Cluster 2.

High order/return volume, low order value, Policy-Abuser-dominated. A shorter return window or restocking fee targets this pattern more precisely than a fraud investigation would.

OWNER · CUSTOMER POLICY

04 Feed the refund-exposure regressor to Finance, not just Fraud & Risk.

Linear Regression prices expected refund dollars the moment a return is filed — a reserve-accounting input independent of whether it turns out to be fraud, abuse, or genuine.

OWNER · FINANCE / CUSTOMER OPS



THANK YOU

Thank you. Questions *welcome.*

We'd rather be asked about the caveats than the number —
\$4.64M of non-legitimate refund exposure, **\$1.88M** of it
reachable today with a single no-ML policy rule.

Subham Ranjan · A017

Shriya · A023

Abhinav Kumar · A035

Dev Vagreacha · A049

Samruddhi Pradhan · A054

Yatharth Vij · A072

IN THE REPOSITORY

notebooks/ — two executable
PySpark notebooks, Bronze to
models

WRITTEN UP

docs/ — per-model results, each
with its own weaknesses section

THE DELIVERABLES

deliverables/ — this deck and the
full consulting report, HTML + PDF

REPRODUCIBILITY

seed=42 throughout · every number
here read from the notebooks' own
output

Dataset: `ecommerce_return_abuse_dataset.csv` (Kaggle) · references, AI-usage disclosure and team contributions in the consulting report's Appendix.